Irradiated Uveal Melanomas: Cytopathologic Correlation With Prognosis

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• PURPOSE: We analyzed the additional prognostic information from cytopathologic data obtained at the time of uveal melanoma irradiation.
• METHODS: Fine-needle aspiration biopsies (FNABs) were performed and rereviewed in a masked manner. These data and standard prognostic variables were correlated with survival and local tumor recurrence using Kaplan-Meier and Cox model statistical techniques.
• RESULTS: Cytopathologic assessment of cell type added to prognostic accuracy. In 116 patients, the percentage of epithelioid cells on FNAB and ciliary body involvement were most strongly correlated with melanoma-related mortality.
• CONCLUSIONS: FNAB data improved our prognostic accuracy in irradiated uveal melanoma patients.

THERE ARE SEVERAL CORRELATES OF UVEAL MELanoma prognosis. Historically, most prognostic variables could only be assessed on histologic examination of enucleated specimens.[1-4] Many eyes with uveal melanoma are now salvaged with radiation therapy, and these pathologic data are not available.[5] In both brachytherapy and charged-particle treatment trials, increased tumor diameter, intraocular melanoma location, presence of localized extraocular extension, and increased patient age have been correlated with a higher incidence of melanoma metastases.[6-9]

Fine-needle aspiration biopsies (FNABs) have been used for intraocular tumor diagnosis for many years.[10-14] We have noted previously an excellent correlation between histologic and cytopathologic assessment of epithelioid cells in enucleated uveal melanomas; in one study,[15] the agreement was greater than 95%. The current prospective investigation was undertaken to assess the prognostic importance of cytopathologic data in uveal melanoma patients treated with either iodine-125 brachytherapy or helium ion irradiation.

PATIENTS AND METHODS

ONE HUNDRED SIXTEEN PATIENTS WERE EXAMINED AND treated in the Ocular Oncology Unit at the University of California, San Francisco. All had given both written and oral permission for entry into these studies. As previously described,[16] all subjects underwent a multimodality, multiobserver evaluation. Uveal melanomas were treated as part of phase I-II or randomized, prospective studies with either helium ion irradiation or iodine-125 brachytherapy.[17]

All patients in this trial had FNABs before treatment. Cytopathologic specimens were obtained at the time of surgical localization for irradiation, as previously described.[18] Most biopsies were performed with a transscleral 25-gauge needle directly over the tumor; the site was isolated, dried, and then closed with glue as the needle was withdrawn. In 12 (10%) of 116 patients, transvitreal pars plana-approach needle biopsies were performed, and tumor specimens were obtained in a similar manner.

The cytopathology specimens were initially examined with an alcohol-fixed rapid stain in the operating room; any specimen not considered a melanoma

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